

Haoran Lei

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EDUCATION

University of Pennsylvania, MSE Electrical Engineering

Philadelphia, PA

- From Sep 2025 to Dec 2026 (Expected)
- GPA: 3.9/4.0
- Relevant Coursework: Smart Devices, Statistics for Data Science, Graph Neural Networks, Nanorobotics, Introduction to Robotics, Feedback Control Design and Analysis, Applied Machine Learning

University of Edinburgh, BEng (Hons) Electronics and Electrical Engineering, First Class

Edinburgh, UK

- From Sep 2021 to Jun 2025
- Final Project: Implementation of a Brain-Computer Interface Model on Raspberry Pi
- Relevant Coursework: Data Analysis and Machine Learning, Digital System Design, Software and Embedded Systems, Signals and Communication Systems, Control and Instrumentation Engineering, Sensors and Instrumentation

WORK EXPERIENCE

Tashan Technology

Beijing, China

Robotics Software Intern / Robotics & Embodied AI Intern

Jun 2026 – Aug 2026

- Developed an RGB-D perception pipeline for a mobile dual-arm manipulation platform by integrating RealSense D435i sensing, YOLO object detection, point-cloud ROI processing, camera calibration, and TF transformations.
- Implemented 3D object-pose estimation and connected perception outputs with MoveIt inverse kinematics and trajectory planning to enable autonomous grasping of bottle-shaped objects.
- Quantized vision models, converted them to RKNN format, and deployed them on an RK3568 edge-computing platform for onboard robotic perception.
- Configured and validated the ROS Noetic and MoveIt stack, including URDF/Xacro/SRDF models, planning groups, joint limits, controllers, and RViz/Gazebo simulation.
- Investigated collision-aware manipulation using Planning Scene, OctoMap, point-cloud updates, and collision objects, and tuned TRAC-IK, KDL, OMPL, and CHOMP parameters.
- Investigated and prototyped vision-language-action and embodied foundation-model approaches for robotic manipulation tasks.
- Diagnosed perception and planning failures by analyzing camera calibration, coordinate-frame transformations, the robot TF tree, and the control pipeline connecting the mobile base, dual arms, cameras, and end effectors.

Lighttech Engineering SL

Elx, Spain

Summer Intern

Aug 2024 – Sep 2024

- Developed and validated embedded control functions for a wireless controller used in automotive and engineering applications, including deadman safety logic, processor interfaces, display integration, charging, USB Power Delivery, and wireless communication.
- Supported hardware architecture evaluation, interface definition, prototype bring-up, and functional debugging while collaborating with mechanical and electrical engineers on design iteration.

Zhongguancun Science City City Brain Co. Ltd

Beijing, China

Computer Vision Intern

Jun 2024 – Aug 2024

- Developed and deployed eight YOLOv5/YOLOv8 computer vision models for human-action recognition and smart-city perception applications.
- Built an end-to-end computer vision workflow covering data collection, annotation, augmentation, model training, validation, deployment, and demonstration.

- Applied vision models to vehicle-color recognition, water-level estimation, tactile-paving obstruction detection, and jumping-action detection, achieving approximately 85% overall accuracy; tuned confidence thresholds and post-processing logic to reduce false detections.

PROJECT EXPERIENCE

Inflatable Soft Wearable Assistive Device, University of Pennsylvania (GRASP Lab, Sung Robotics Lab)

Philadelphia, PA

Master's Thesis Project and Lead Student Researcher at Penn

Nov 2025 – Present

- Lead the development of sensing and control software for an inflatable wearable assistive device in collaboration with Stanford University's CHARM Laboratory.
- Develop an IMU-based motion-processing pipeline to track body posture and distinguish normal movement from loss-of-balance conditions in real time.
- Implement closed-loop pressure-control logic that processes pressure-sensor feedback and regulates pneumatic chamber inflation to maintain target pressure.
- Develop solenoid-valve control logic that coordinates posture estimation, pressure feedback, and pneumatic actuation for real-time assistance.
- Build sensor-calibration, data-logging, parameter-tracking, and repeatable testing workflows to evaluate control stability, compliance, and system repeatability.

Implementation of a Brain-Computer Interface Model on Raspberry Pi

Edinburgh, UK

Bachelor's Thesis Project

Sep 2024 – Jun 2025

- Developed a non-invasive brain-computer interface using EEG and EOG signals for motor-imagery classification.
- Built a signal-processing and machine-learning pipeline covering filtering, normalization, segmentation, model training, and evaluation.
- Trained and evaluated RNN and LSTM models, achieving approximately 96% validation accuracy with stable AUC performance.
- Deployed real-time machine-learning inference on a Raspberry Pi 5 and optimized latency, memory usage, and thermal constraints for edge-robotics and portable-system applications.
- Designed a GPIO switch and LED PCB that integrated embedded hardware with the inference and control pipeline.
- Compared preprocessing windows, model architectures, and evaluation metrics, analyzed misclassified samples, and improved the interpretability of the embedded inference workflow.

HypED – Student Technology Society Developing Hyperloop Pod

Edinburgh, UK

Electronics Navigation Project Manager

Sep 2023 – Aug 2024

- Integrated optical-flow, full-spectrum, and acceleration sensors with a Raspberry Pi 4 for pod-speed estimation and wireless control.
- Designed and validated a navigation PCB incorporating Raspberry Pi interfaces, an accelerometer, voltage regulation, and multiple sensor connections.
- Coordinated electronics debugging, hardware validation, and subsystem integration and developed troubleshooting procedures for sensor drift, power stability, and wireless-link reliability.

PROFESSIONAL SUMMARY

Electrical engineering graduate student focused on robotics software, embodied AI, embedded sensing, controls, and ML deployment. Built hardware-software systems across mobile manipulation, wearable robotics, BCI, hyperloop navigation, computer vision and VLA.

SKILLS

Programming and Development: Python, C, C++, Verilog, Bash, Linux, Git

Robotics and Control: ROS Noetic, MoveIt, RViz, Gazebo, URDF/Xacro/SRDF, TF, ros_control, inverse kinematics, motion planning, grasp planning, feedback control

Perception and Machine Learning: YOLOv5/YOLOv8/YOLO26, PyTorch, RGB-D perception, point-cloud processing, camera calibration, RNN, LSTM, GNN, EEG/EOG signal processing
Embedded and Hardware: Raspberry Pi 4/5, RK3568/RK3588, RKNN deployment, RealSense D435i, GPIO integration, Altium Designer, LTspice, Vivado, Xilinx SDK

LANGUAGES: Mandarin Chinese (Native), English (Fluent), Spanish (B2)